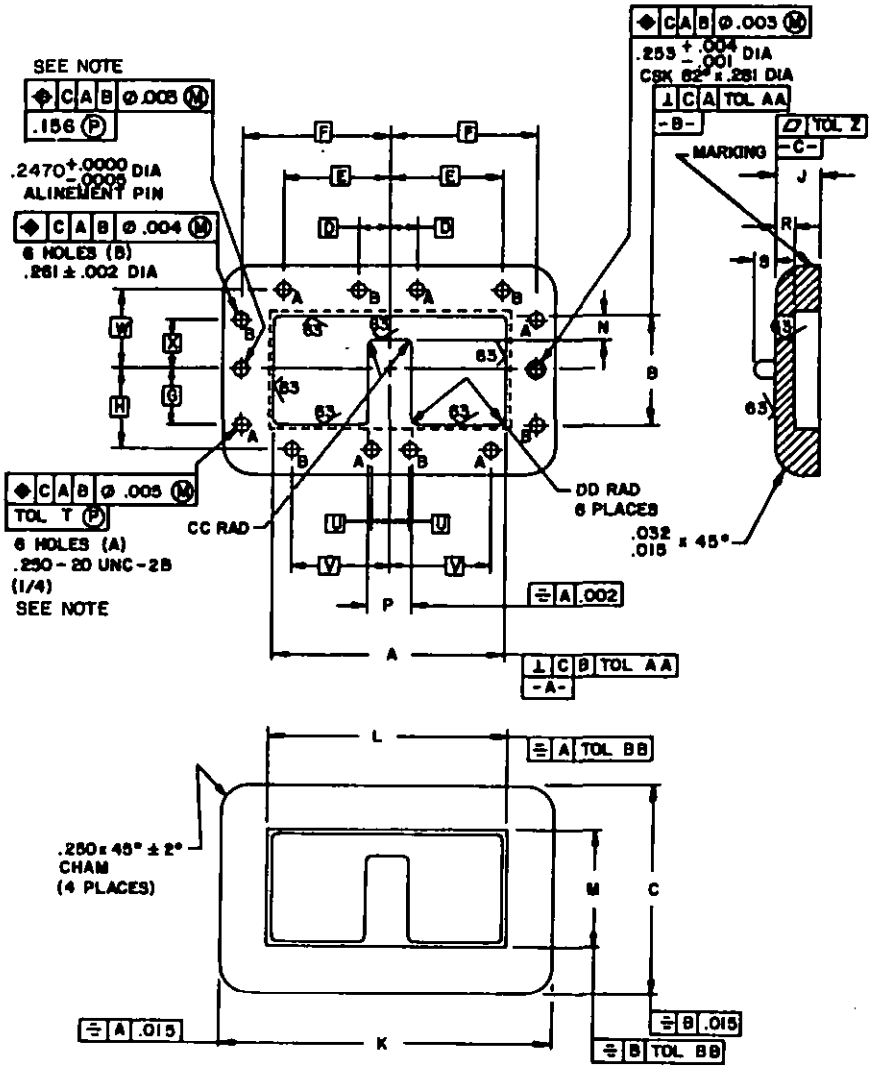
MILITARY SPECIFICATION SHEET

FLANGES, WAVEGUIDE, SINGLE RIDGE, SOCKET MOUNT (BANDWIDTH RATIO 2.4:1)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the flanges described herein shall consist of this document and the latest issue of Specification MIL-F-39000.



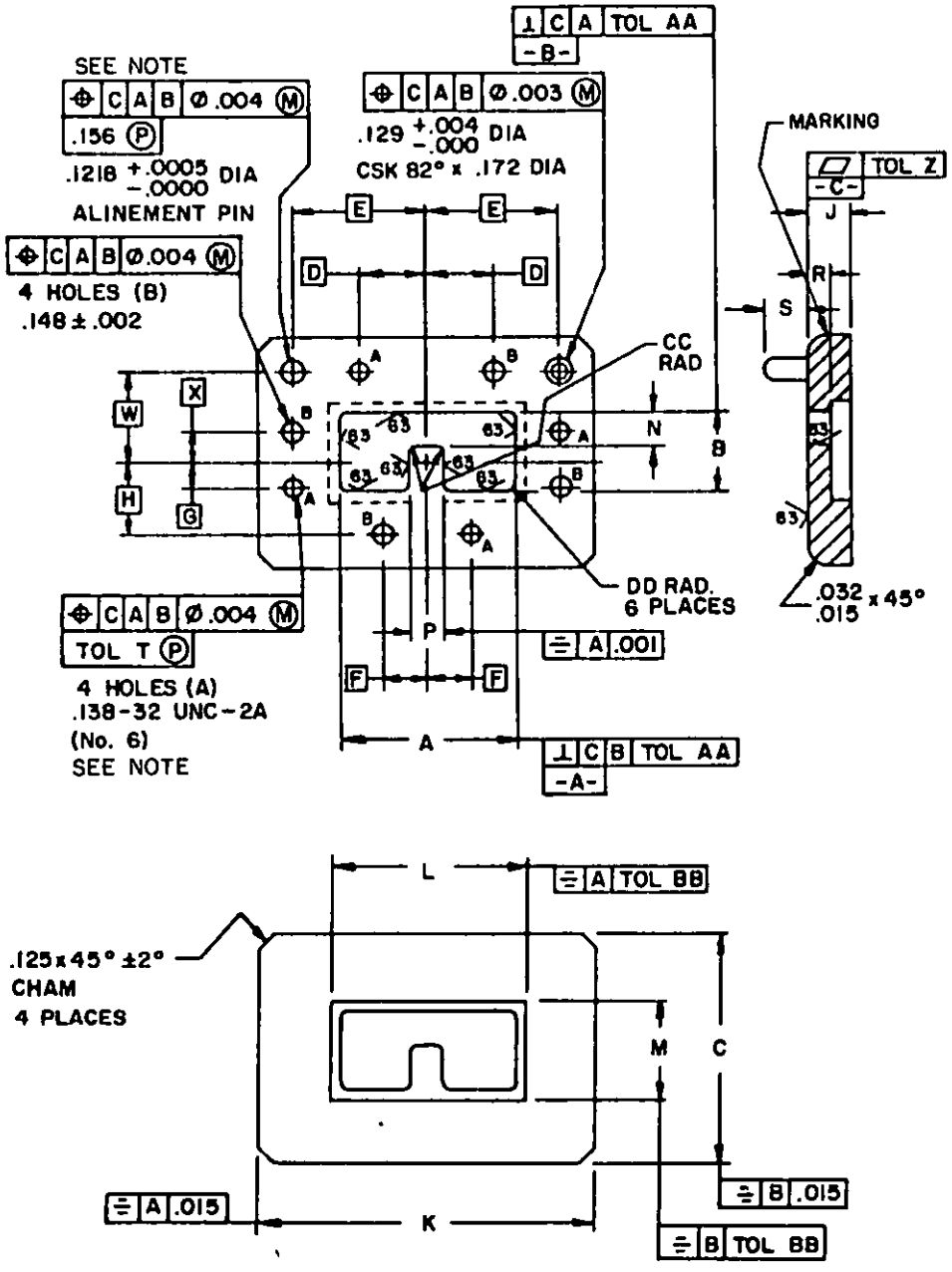
Configuration 1.

FIGURE 1. Flange.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, Department of the Navy, ATTN: ELEX 5043, Washington, D.C. 20360 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5985

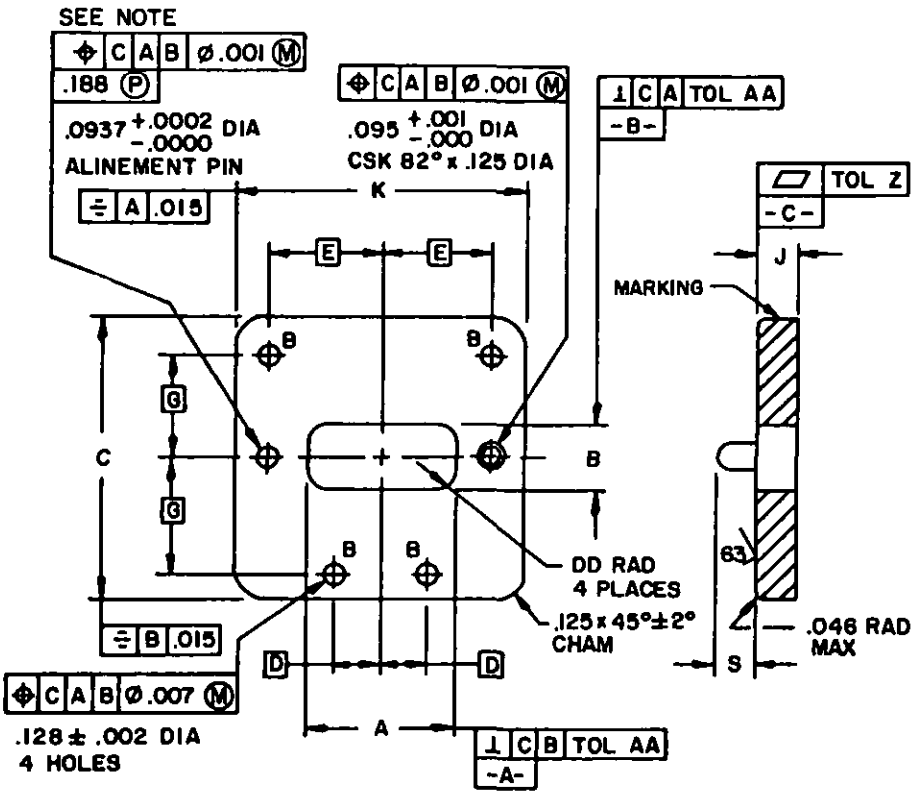
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Configuration 2.

FIGURE 1. Flange - Continued.

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Configuration 3.

INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.0002	.005	.005	.13	.1218	3.094	.172	4.37
.0005	.013	.007	.18	.125	3.18	.247	6.27
.001	.02	.015	.38	.128	3.25	.250	6.35
.002	.05	.032	.81	.129	3.28	.253	6.43
.003	.08	.046	1.17	.138	3.51	.261	6.63
.004	.10	.0937	2.380	.148	3.76	.281	7.14
		.095	2.41	.156	3.96		

NOTE: The projected tolerance zone surface reference is datum C.

FIGURE 1. Flange - Continued.

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TABLE I. Part number characteristics and dimensions.

Part number M39000/1-			Part number M39000/1-			Configuration	Dimensions 1/ 2/ 3/	
	Used with			Used with				
Dash No. 4/	Waveguide M23351/3-	Mating flange M39000/1-	Dash No. 5/	Waveguide M23351/3-	Mating flange M39000/1-		A	B
001	017	001	002	018, 019, 020	002	1	5.847 ±.005 (148.51) (.13)	2.631 ±.005 (66.83) (.13)
004	021	004	005	022, 023, 024	005	1	3.276 ±.005 (83.21) (.13)	1.474 ±.005 (37.44) (.13)
007	025	007	008	026, 027, 028	008	2	2.456 ±.004 (62.38) (.10)	1.105 ±.004 (28.07) (.10)
010	029	010	011	030, 031, 032	011	2	1.404 ±.003 (35.66) (.08)	.632 ±.003 (16.05) (.08)
013	033	013	014	034, 035, 036	014	2	1.034 ±.003 (26.26) (.08)	.465 ±.003 (11.81) (.08)
016	037	016	017	038, 039, 040	017	3	.7655 ±.0020 (19.444) (.051)	.4055 ±.0020 (10.300) (.051)
019	041	019	020	042, 043, 044	020	3	.537 ±.002 (13.64) (.05)	.291 ±.002 (7.39) (.05)
022	045	022	023	046, 047, 048	023	3	.362 ±.002 (9.19) (.05)	.212 ±.002 (5.38) (.05)

Dash No. 4/	Dash No. 5/	Dimensions 1/ 2/ 3/								
		C ±.015 (.38)	D BSC.	E BSC.	F BSC.	G BSC.	H BSC.	J ±.015 (.38)	K ±.015 (.38)	L
001	002	4.656 (118.26)	.922 (23.42)	1.844 (46.84)	3.391 (86.13)	---	---	.500 (12.70)	7.875 (200.02)	6.023 ±.005 (152.98) (.13)
004	005	3.625 (92.08)	.578 (14.68)	1.734 (44.04)	2.172 (55.17)	---	---	.500 (12.70)	5.250 (133.35)	3.450 ±.004 (87.63) (.10)
007	008	2.062 (52.37)	.763 (19.38)	1.513 (38.43)	.635 (16.13)	---	---	.438 (11.13)	3.438 (87.33)	2.629 ±.003 (66.78) (.08)
010	011	1.562 (39.67)	.399 (10.13)	.961 (24.41)	.271 (6.88)	---	---	.438 (11.13)	2.312 (58.72)	1.544 ±.003 (39.22) (.08)
013	014	1.312 (33.32)	.377 (9.58)	.752 (19.10)	.249 (6.32)	---	---	.250 (6.35)	1.875 (47.62)	1.145 ±.002 (29.08) (.05)
016	017	1.375 (34.92)	.429 (10.90)	.530 (13.46)	---	.500 (12.70)	---	.188 (4.78)	1.375 (34.92)	---
019	020	1.000 (25.40)	.306 (7.77)	.388 (9.86)	---	.344 (8.74)	---	.188 (4.78)	1.000 (25.40)	---
022	023	.875 (22.22)	.224 (5.69)	.312 (7.92)	---	.280 (7.11)	---	.188 (4.78)	.875 (22.22)	---

See footnotes at end of table.

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TABLE I. Part number characteristics and dimensions - Continued.

Dash No. 4/	Dash No. 5/	Dimensions 1/ 2/ 3/						
		M	N	P	R ±.005 (.13)	S ±.005 (.13)	T	U BSC.
001	002	2.807 ±.005 (71.30) (.13)	1.097 ±.003 (27.86) (.08)	.9060 ±.0025 (23.012) (.064)	.375 (9.52)	.156 (3.96)	.500 (12.70)	.672 (17.07)
004	005	1.648 ±.004 (41.86) (.10)	.6150 ±.0025 (15.621) (.064)	.508 ±.002 (12.90) (.05)	.375 (9.52)		.500 (12.70)	.328 (8.33)
007	008	1.278 ±.003 (32.46) (.08)	.461 ±.002 (11.71) (.05)	.381 ±.002 (9.68) (.05)	.250 (6.35)		.438 (11.13)	---
010	011	.772 ±.003 (19.61) (.08)	.2640 ±.0015 (6.706) (.038)	.2180 ±.0015 (5.537) (.038)	.250 (6.35)		.438 (11.13)	---
013	014	.576 ±.002 (14.63) (.05)	.194 ±.001 (4.93) (.03)	.160 ±.001 (4.06) (.03)	.125 (3.18)		.250 (6.35)	---
016	017	---	---	---	---		---	---
019	020	---	---	---	---		---	---
022	023	---	---	---	---	.156 (3.96)	---	---

Dash No. 4/	Dash No. 5/	Dimensions 1/ 2/ 3/							
		V BSC.	W	X	Z	AA	BB	CC	DD Max.
001	002	2.516 (63.91)	---	---	.0007 (.018)	.005 (.13)	.005 (.13)	.219 ±.020 (5.56) (.51)	.047 (1.19)
004	005	1.484 (37.69)	---	---	.0007 (.018)	.005 (.13)	.004 (.10)	.123 ±.012 (3.12) (.30)	.047 (1.19)
007	008	---	---	---	.0007 (.018)	.004 (.10)	.003 (.08)	.092 ±.009 (2.34) (.23)	.047 (1.19)
010	011	---	---	---	.0004 (.010)	.003 (.08)	.003 (.08)	.053 ±.005 (1.32) (.13)	.031 (.79)
013	014	---	---	---	.0004 (.010)	.003 (.08)	.002 (.05)	.039 ±.003 (.99) (.08)	.031 (.79)
016	017	---	---	---	.0003 (.008)	.002 (.05)	---	---	.070 (1.78)
019	020	---	---	---	.0003 (.008)	.002 (.05)	---	---	.060 (1.52)
022	023	---	---	---	.0003 (.008)	.002 (.05)	---	---	.060 (1.52)

1/ Dimensions are in inches.
2/ Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3/ Metric equivalents are in parentheses.
4/ Material shall be aluminum alloy.
5/ Material shall be copper alloy.

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REQUIREMENTS:

Dimensions and configuration: See figure 1 and table I.

Material: Metallic alloy as specified in table I.
Materials supplied with flanges: See table II.

Marking:
Part number: M39000/1-(dash number from table I).
Location: See figure 1.

Cross reference and frequency range: See table III.

TABLE II. Items supplied with flanges.

Flange part No. M39000/1-	Socket head cap screw	Hex nut	Spring lockwasher
001, 002, 004, 005	.250-20 UNC-2A 1.25 in. long (6 ea)	—	.250 x .062 (6 ea)
007, 008, 010, 011	.138-32 UNC-2A 1.0 in. long (4 ea)	—	.138 x .031 (4 ea)
013, 014	.138-32 UNC-2A .625 in. long (4 ea)	—	.138 x .031 (4 ea)
016, 017, 019, 020, 022, 023	.112-40 UNC-2A .50 in. long (2 ea)	.112-40 UNC-2B (2 ea)	.112 x .025 (2 ea)

TABLE III. Cross reference and frequency range.

Part number 1/ M39000/1-	AN nomenclature	Frequency range (GHz)
001	UG-1541/U	.84- 2.00
002	UG-1542/U	.84- 2.00
004	UG-1544/U	1.50- 3.60
005	UG-1545/U	1.50- 3.60
007	UG-1547/U	2.00- 4.80
008	UG-1548/U	2.00- 4.80
010	UG-1550/U	3.50- 8.20
011	UG-1551/U	3.50- 8.20
013	UG-1553/U	4.75-11.00
014	UG-1554/U	4.75-11.00
016	UG-1556/U	7.50-18.00
017	UG-1557/U	7.50-18.00
019	UG-1559/U	11.00-26.50
020	UG-1560/U	11.00-26.50
022	UG-1562/U	18.00-40.00
023	UG-1563/U	18.00-40.00

1/ Dash numbers 003 (UG-1543/U), 006 (UG-1546/U), 009 (UG-1549/U), 012 (UG-1552/U), 015 (UG-1555/U), 018 (UG-1558/U), 021 (UG-1561/U), and 024 (UG-1564/U) have been deleted from specifications.

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Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army -
Navy - SH
Air Force - 11, 99
DLA - ES

User activities:
Army - AV, AR
Navy - MC, AS, OS, SH, CG
Air Force - 14, 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

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